

One Pager on QuantumTrace

Cryptographic Agility & Post-Quantum Readiness Scanner

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The Challenge

Challenge name/tagline: QuantumTrace – Uncovering Cryptographic Debt in the Quantum Dawn.

Challenge one-liner: Build an automated discovery and assessment engine to detect, classify, and visualize quantum-vulnerable cryptographic algorithms across network traffic, application code, and cloud environments.

Introduction

The emergence of cryptanalytically relevant quantum computers poses an existential threat to modern digital trust. Fundamental public-key algorithms underlying the internet – such as RSA, Elliptic Curve Cryptography (ECDSA, ECDH), and Diffie-Hellman – are fully breakable using Shor's Algorithm. Additionally, adversaries are already actively executing "*Harvest Now, Decrypt Later*" (HNDL) attacks: intercepting and archiving encrypted enterprise communications today to decrypt them once quantum systems mature.

While standard bodies have finalized post-quantum standards (such as NIST FIPS 203, 204, and 205), most organizations remain entirely blind to where legacy cryptographic primitives are actively utilized across their environments. Enterprise security teams lack real-time visibility into their cryptographic footprint, making remediation, cryptographic agility, and compliance roadmaps almost impossible to plan or execute.

This challenge invites participants to develop QuantumTrace, a lightweight, non-intrusive discovery tool designed to uncover classical cryptographic assets, map vulnerabilities against quantum threats, and generate actionable Cryptography Bill of Materials (CBOM).

The Core of the Challenge

The goal is to build an automated engine that audits heterogeneous inputs to pinpoint cryptographic mechanisms vulnerable to Shor's and Grover's algorithms. To ensure rapid development, participants can focus on one or more of the following vectors using provided synthetic data samples:

1. Network Layer Discovery (In-Transit):

- Parse captured network sessions to analyze TLS handshakes (Client/Server Hello).
- Identify negotiated cipher suites, supported key exchange groups, and signature schemes, distinguishing between classical algorithms (e.g., RSA, secp256r1) and hybrid/post-quantum primitives (e.g., ML-KEM/Kyber).

2. Application & Code Inspection (At-Rest / In-Use):

- Scan provided mock codebases, configuration files, or binary artifacts to extract cryptographic references.
- Generate a normalized Cryptography Bill of Materials (CBOM) representing declared algorithms, key lengths, and cryptographic libraries.

3. Quantum Risk Classification Engine:

- Automatically categorize identified primitives into clear risk tiers:
- Critical: Asymmetric schemes vulnerable to Shor's algorithm (RSA, DSA, ECDSA, ECDH).
- Medium: Symmetric ciphers/hashes weakened by Grover's algorithm (e.g., AES-128, 3DES, SHA-1).
- Safe / Quantum-Resilient: Large-key symmetric primitives (AES-256) and recognized Post-Quantum Cryptography (PQC) standards.

4. Actionable Dashboard & Reporting:

- Provide a clean interface (CLI or Web UI) visualizing the enterprise "Quantum Risk Score" and exporting a structured audit report (JSON/CycloneDX format).

Provided Starter Kit & Data Sources

To ensure no direct access to live corporate infrastructure is required, the challenge organizers will provide:

- Synthetic network traffic captures (.pcap / .pcapng files) covering legacy TLS 1.2 sessions, modern TLS 1.3 handshakes, and hybrid PQC exchanges.
- Sample code repositories (Python, Java, Go) containing intentional classical and modern cryptographic patterns.
- Mock Cloud Key Management Service (KMS) and Load Balancer configuration dumps (JSON format).
- Optional live sandbox targets: 2–3 public endpoints exposing different TLS cipher suite profiles for real-time probing.

Suggested Technical Components

- Backend & Scripting: Python (Scapy, PyShark, cryptography), Go, or Node.js.
- Network Analysis: TShark/Wireshark libraries, Scapy, or native raw socket parsers.
- CBOM & Schemas: CycloneDX 1.6 (Crypto asset module) or custom JSON schemas.
- Frontend / Visualization: Streamlit, Gradio, React, or terminal-based UI (Rich/Textual).

Evaluation Criteria

Solutions will be evaluated based on the following criteria:

1. Detection Accuracy & Risk Scoring (40%): Correct identification of cryptographic primitives, accurate differentiation between Shor-vulnerable and Grover-weakened algorithms, and minimal false classifications.
2. Data Ingestion & Coverage (25%): Ability to process multiple input types (PCAP traces, code snippets, or configuration dumps).
3. Standardization & Traceability (20%): Quality of the generated output (e.g., CBOM format) and clear mapping to recognized PQC standards (NIST FIPS 203/204/205).
4. Usability & Demonstration (15%): Clarity of the dashboard/CLI interface, ease of local setup (e.g., Dockerfile/single script), and effectiveness of the final demo pitch.

Future Potential and Benefits

- QuantumTrace bridges the gap between academic quantum theory and practical enterprise cyber defense. By equipping organizations with automated cryptographic observability, it enables:
- Defending against current Harvest Now, Decrypt Later interception strategies.
- Accelerating organizational cryptographic agility ahead of regulatory PQC enforcement deadlines.
- Establishing foundational automation for enterprise-wide CBOM generation and continuous security compliance.